

We prove that the maximum index is $N = 5$, the smallest possible maximum element for such a multiset is $M = 16$, and among all multisets with these N and M the minimum number of elements is 32.

1. Generating function formulation For a multiset S with multiplicities $f(k)$ ($1 \leq k \leq 19$) define

$$P(z) = \sum_{k=1}^{19} f(k)z^k.$$

The condition “ S can be partitioned into multisets A_i and B_i with $A_i = i + B_i$ ” is equivalent to the existence of a polynomial $Q_i(z)$ with non-negative integer coefficients such that

$$P(z) = (1 + z^i)Q_i(z).$$

(Indeed, writing $Q_i(z) = \sum_t g_i(t)z^t$ where $g_i(t)$ is the multiplicity of t in B_i , the equality holds and Q_i automatically has degree at most $19 - i$.) Hence the index of S is the largest n for which $P(z)$ is divisible by $1 + z^1, \dots, 1 + z^n$ and each quotient has non-negative coefficients.

2. Cyclotomic factorisation and the LCM For every i ,

$$1 + z^i = \prod_{\substack{d|2i \\ d \nmid i}} \Phi_d(z),$$

where Φ_d is the d -th cyclotomic polynomial. The set of d that occur for all $1 \leq i \leq n$ is exactly the set of even integers d with $2 \leq d \leq 2n$. Consequently the least common multiple of $1 + z^1, \dots, 1 + z^n$ is

$$L_n(z) = \prod_{\substack{2 \leq d \leq 2n \\ d \text{ even}}} \Phi_d(z).$$

Its degree is $\sum_{k=1}^n \varphi(2k)$. For $n = 6$ this degree is $\varphi(2) + \varphi(4) + \varphi(6) + \varphi(8) + \varphi(10) + \varphi(12) = 1 + 2 + 2 + 4 + 4 + 4 = 17$; for $n = 7$ it is $23 > 19$. Since $P(z)$ must be a multiple of $L_n(z)$ and $\deg P \leq 19$, we must have $\deg L_n \leq 19$; therefore $n \leq 6$.

3. Impossibility for $n = 6$ If an S had index 6, then $P(z) = L_6(z)R(z)$ with $R(0) = 0$ and $\deg R \leq 2$ (because $\deg L_6 = 17$). Thus $R(z) = r_1z + r_2z^2$ with integers r_1, r_2 . Write $M_1(z) = L_6(z)/(1 + z)$; then

$$Q_1(z) = \frac{P(z)}{1 + z} = M_1(z)(r_1z + r_2z^2)$$

must have non-negative coefficients. The coefficients of $M_1(z)$ (obtained by dividing L_6 by $1 + z$) are

$$M_1(z) = 1 - 2z + 3z^2 - 3z^3 + 4z^4 - 4z^5 + 5z^6 - 5z^7 + 6z^8 - 5z^9 + 5z^{10} - 4z^{11} + 4z^{12} - 3z^{13} + 3z^{14} - 2z^{15} + z^{16}.$$

The first few coefficients of Q_1 give

$$q_1 = r_1, \quad q_2 = -2r_1 + r_2, \quad q_{14} = r_1 - 2r_2, \quad \dots$$

The conditions $q_2 \geq 0$ and $q_{14} \geq 0$ force $r_2 \geq 2r_1$ and $r_2 \leq r_1/2$, which imply $r_1 = r_2 = 0$. Hence no non-zero solution exists, so index 6 is impossible. Therefore $N \leq 5$.

4. Construction for $n = 5$ Take

$$P(z) = z \prod_{i=1}^5 (1 + z^i).$$

This polynomial has degree 16 and all coefficients are non-negative (they count partitions into distinct parts). For each $i = 1, \dots, 5$ we have

$$\frac{P(z)}{1+z^i} = z \prod_{\substack{j=1 \\ j \neq i}}^5 (1+z^j),$$

which also has non-negative coefficients. Hence the corresponding multiset S has index at least 5. By the previous part it cannot have index 6, so index exactly 5. Consequently the maximum possible index is $N = 5$.

5. Determining the minimal maximum element M Let S be a multiset of index 5. Then $P(z)$ is a multiple of the LCM $L_5(z) = \prod_{d=2,4,6,8,10} \Phi_d(z)$, which has degree 13; so $P(z) = L_5(z)R(z)$ with $R(0) = 0$. The largest element of S is $\deg P$. If the maximum element were ≤ 15 , then $\deg P \leq 15$ and consequently $\deg R \leq 2$.

- For $\deg R = 1$: $R(z) = r_1 z$. Then $P(z) = r_1 z L_5(z)$; but L_5 has negative coefficients (e.g. the coefficient of z^1 is -1), so P would have a negative coefficient – impossible.
- For $\deg R = 2$: $R(z) = r_1 z + r_2 z^2$. The non-negativity of the coefficients of P forces $r_1 = r_2 > 0$. Then

$$Q_1(z) = \frac{P(z)}{1+z} = r_1 z L_5(z),$$

and $L_5(z)$ contains the term $-z$, so Q_1 contains a negative term – contradicting the index condition for $i = 1$. Hence the maximum element cannot be ≤ 15 . The construction in step 4 gives a multiset with maximum element 16 and index 5, so the smallest possible maximum element is $M = 16$.

6. Minimising the size for index 5 and maximum 16 For a multiset S with index 5 and maximum 16 we have $\deg P = 16$, so $R(z)$ has degree 3. Write $R(z) = r_1 z + r_2 z^2 + r_3 z^3$ with integers $r_1, r_2, r_3 \geq 0$ (the coefficients of P must be non-negative; from the expressions of the coefficients one easily deduces $r_1, r_2, r_3 \geq 0$). The total number of elements is $|S| = P(1) = L_5(1) \cdot R(1)$. Evaluating the cyclotomic polynomials at 1 gives

$$\Phi_2(1) = 2, \quad \Phi_4(1) = 2, \quad \Phi_6(1) = 1, \quad \Phi_8(1) = 2, \quad \Phi_{10}(1) = 1,$$

so $L_5(1) = 8$. Hence $|S| = 8(r_1 + r_2 + r_3)$. Now consider the condition for $i = 1$. Write $M_1(z) = L_5(z)/(1+z)$ (the polynomial obtained by dividing L_5 by $1+z$); its coefficients are

$$M_1(z) = 1 - 2z + 4z^2 - 5z^3 + 7z^4 - 7z^5 + 8z^6 - 7z^7 + 7z^8 - 5z^9 + 4z^{10} - 2z^{11} + z^{12}.$$

The requirement that $Q_1(z) = M_1(z)R(z)$ has non-negative coefficients yields, in particular,

$$(i) \quad \text{coefficient of } z^2 : -2r_1 + r_2 \geq 0 \Rightarrow r_2 \geq 2r_1,$$

$$(ii) \quad \text{coefficient of } z^{14} : r_2 - 2r_3 \geq 0 \Rightarrow r_2 \geq 2r_3,$$

and also $r_1 > 0$ (otherwise a contradiction with later inequalities) and $r_3 > 0$ (because the coefficient of z^{16} in P equals r_3 , and the maximum element is 16). Therefore $r_1 \geq 1$, $r_3 \geq 1$ and $r_2 \geq 2$, so $r_1 + r_2 + r_3 \geq 1 + 2 + 1 = 4$. Consequently $|S| \geq 8 \cdot 4 = 32$. The equality is attained by $(r_1, r_2, r_3) = (1, 2, 1)$, i.e. $R(z) = z(1+z)^2$, which gives the polynomial of the construction in step 4. This choice indeed satisfies all the required non-negativity conditions (all quotients Q_i are products of factors $1+z^j$ with a factor z , hence have non-negative coefficients). Thus the minimal possible number of elements is 32.

Hence the answer is $\boxed{32}$.